



Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir
M.V. College
Of Science & Commerce

Department of Information Technology (Bsc.IT Semester III)
SCALA FOR D.S PRACTICAL 26

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PRACTICAL 26

Question:

Perform logistic regression using Breeze. Classify a dataset with binary labels.

CODE:

```

1  import breeze.linalg._
2
3  object Demo extends App {
4
5      val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0) // Dataset
6
7      val y = DenseVector(0.0, 0.0, 0.0, 1.0, 1.0) // Binary labels
8
9      // Initial weight and bias
10     var weight = 0.0
11     var bias = 0.0
12
13     // Sigmoid function
14     def sigmoid(z: Double): Double = {
15         1.0 / (1.0 + math.exp(-z))
16     }
17
18     // Train the model
19     for (i <- 1 to 1000) {
20         for (j <- 0 until x.length) {
21
22             val prediction = sigmoid(weight * x(j) + bias)
23             val error = prediction - y(j)
24
25             weight = weight - 0.01 * error * x(j)
26             bias = bias - 0.01 * error
27         }
28     }
29
30     // Classify the data
31     println("Predictions:")
32
33     for (i <- 0 until x.length) {
34
35         val probability = sigmoid(weight * x(i) + bias)
36

```

```

// Classify the data
println("Predictions:")

for (i <- 0 until x.length) {

    val probability = sigmoid(weight * x(i) + bias)

    val result =
        if (probability >= 0.5) 1 else 0

    println(
        "Input = " + x(i) +
        ", Probability = " + probability +
        ", Class = " + result
    )
}
println("Rahul_51")
}

```



The screenshot shows a code editor with two tabs: 'build.sbt' and 'Demo.scala'. The 'build.sbt' tab is active, displaying the following code:

```

1 ThisBuild / version := "0.1.0-SNAPSHOT"
2
3 ThisBuild / scalaVersion := "2.12.18"
4 libraryDependencies += "org.scalanlp" %% "breeze" % "2.1.0"
5 lazy val root = (project in file("."))
6   .settings(
7     name := "Prac25"
8   )
9

```

OUTPUT:

```
Run Demo x
" C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
Predictions:
Input = 1.0, Probability = 0.0515501057576029, Class = 0
Input = 2.0, Probability = 0.16412826954644827, Class = 0
Input = 3.0, Probability = 0.4149888799701166, Class = 0
Input = 4.0, Probability = 0.7193154698480789, Class = 1
Input = 5.0, Probability = 0.9025176083649726, Class = 1
Rahul_51

Process finished with exit code 0
```